Towards a mechanistic understanding of the drivers of plant-pollinator interactions

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Abstract

Flowering plants exhibit an astonishing diversity of reproductive structures, which shapes how they interact with their pollinators. However, to what extent reproductive traits or other drivers like phenology and species abundances shape plant-pollinator interactions is still an open question. To disentangle the relative effects of species phenology, abundances and traits in plant-pollinator interactions, we compared pollinator visitation rates, diet breath, specialization from plant species that vary in reproductive traits, phenology and abundances. To do this, we observed plant-pollinator interactions for a whole flowering season in three different botanical gardens located in an urban setting. We anticipate that the ecological role of the different plant and pollinator species are largely driven by their phenological overlap and abundance, and only certain taxonomic groups are strongly influenced by trait matching processes. Our results highlight the relevance of considering multiple mechanisms to understand plant-pollinator interactions.

Introduction

Flowering plants, pollinators, and their interactions form complex networks that are fundamental to the architecture of biodiversity. The structure and composition of these networks exhibit strong spatio-temporal variation, shaped by species attributes and environmental conditions (Burkle & Alarcón, 2011; Tylianakis & Morris, 2017). While ecological networks provide insights into the ecological processes that govern community assembly (Bascompte & Jordano, 2007; Guimaraes Jr, 2020), documenting interactions remains extremely labour-intensive (Chacoff et al., 2012). Moreover, the resulting network structures are also highly dependent on sampling effort and methodology (Jordano, 2016). At the same time, interactions are often summarised into single metrics and analysed with a diverse array of statistical methods, each offering potentially distinct perspectives on the ecological system (Delmas et al., 2019; Peralta et al., 2024). Despite pronounced temporal dynamics, networks are often treated as static entities (CaraDonna et al., 2017; Poisot et al., 2015), requiring the aggregation or simplification of relevant information when evaluating ecological patterns. The complexity of plant-pollinator networks, together with the limitations of current analytical practices, can obscure rather than clarify our view of ecological systems. Addressing these challenges requires systematic approaches that integrate multiple layers of network complexity to improve both understanding and predictive power (Blüthgen & Staab, 2024; Dormann et al., 2017).

There is strong evidence that species phenology, abundances, and traits are key ecological drivers shaping species interactions and network structure (Kaiser-Bunbury et al., 2014; Vázquez et al., 2009), yet most studies lack the appropriate data to assess their combined influence (Olito & Fox, 2015; Vázquez et al., 2009). Phenology is considered one of the main determinants of network structure, as temporal overlap between species constrains the pool of potential interactions (Benadi et al., 2014; Encinas-Viso et al., 2012), with approximately one-third of all potential interactions estimated to be impossible due to phenological mismatches (Olesen et al., 2011). When two species overlap in time, the probability that they interact is generally considered a function of their relative abundances and species traits (Olito & Fox, 2015; Stang et al., 2007), along with other abiotic and biotic factors (e.g. temperature and species competition). While abundance influences both species' roles (Fort et al., 2016; Simmons et al., 2019) and overall network structure (Canard et al., 2014; Vázquez et al., 2009), independent estimates are rarely available and are often inferred from interaction data (Blüthgen & Staab, 2024).

Traits, in turn, provide mechanistic insights, as they constrain the range of possible partners and help define species' ecological roles within networks (Bartomeus et al., 2016; Coux et al., 2016; Dehling et al., 2016). For instance, interaction probability often peaks when pollinator proboscis length matches floral tube depth (Klumpers et al., 2019), yet imperfect matches can still allow interactions. Although multiple traits often act simultaneously, capturing their multidimensional nature remains challenging (Eklöf et al., 2013; Lanuza et al., 2023), though phylogenetic approaches may help integrate such complexity. Further, most pollinators tend to be generalist, and the relative importance of traits is expected to decline with increasing generalism (Peralta et al., 2020). The strong context-dependency of these different ecological drivers constrains predictive power and highlights the need for high-quality data to evaluate multiple scenarios across taxonomic groups and spatiotemporal scales.

Recent evidence suggests that distinct temporal scales can provide different views of the structure and processes that shape plant-pollinator interactions (Miele et al., 2020; Schwarz et al., 2020). Species and interactions are highly dynamic, shifting markedly between years (Chacoff et al., 2018; Petanidou et al., 2008), across seasons (Souza et al., 2018; Vizentin-Bugoni et al., 2016), and even within days (Schwarz et al., 2021). At fine scales, interactions are shaped by physiological responses to abiotic factors (e.g., temperature, light, precipitation) that determine activity periods, whereas variation across days or weeks also reflects phenological succession and changes in the species pool (Bramon Mora et al., 2020; CaraDonna et al., 2017). Although some of this variation arises from natural dynamics, global change is expected to disrupt and intensify temporal fluctuations (Duchenne et al., 2020; Kharouba et al., 2018). While aggregated snapshots of communities are valuable for describing structural and ecological properties at broad scales (Bascompte & Jordano, 2007), the relative impact of ecological drivers may vary with temporal scale, underscoring the need to account for temporal dynamics to fully understand plant-pollinator interactions.

Although there is no consensus on the range of tools used to analyze ecological networks, combining approaches can provide complementary insights (Blüthgen & Staab, 2024; Delmas et al., 2019; Dormann et al., 2025). Structural metrics are often compared with synthetic networks or null models built under different ecological constraints to test for significance, or across empirical networks to assess variation among systems (Marjakangas et al., 2022; Santamaria & Rodriguez-Girones, 2007; Vázquez & Aizen, 2004), a strategy widely applied to evaluate properties such as nestedness (Bascompte et al., 2003; Payrató-Borras et al., 2019; Ulrich & Gotelli, 2007) and modularity (Dormann et al., 2013; Olesen et al., 2007). Other statistical approaches can complement structural analyses by explicitly assessing the influence of biotic and abiotic drivers. Common approaches include matrix comparisons such as Mantel or Procrustes tests (Burkle & Alarcón, 2011; Dehling et al., 2016) and conventional models such as generalized linear models (Classen et al., 2020; Fornoff et al., 2017; Herrera, 2020), which allow more refined evaluations of the impact of biotic and abiotic drivers on species interactions. In addition, machine learning approaches provide more flexible alternatives that can capture nonlinear relationships and often outperform conventional methods. (Pichler et al., 2020). Despite their complementary nature, selecting and interpreting the outputs of these different statistical tools remains challenging and requires a careful evaluation of their respective strengths and limitations when addressing ecological questions.

Predicting species interactions is a central goal in ecology (Peralta et al., 2024; Valdovinos, 2019), particularly in the context of climate change, ongoing species loss, and the need to sustain ecosystem services. However, although overall network structure can be partially reproduced (Crea et al., 2016; Eklöf et al., 2013), accurately predicting pairwise interactions remains a major challenge (Peralta et al., 2024; Valdovinos, 2019). Recent efforts to improve predictive performance have sought to address limitations of conventional analytical methods, such as non-independence of observations and non-linear effects of key variables, while incorporating the aforementioned ecological drivers (Bartomeus et al., 2016; Benadi et al., 2022). In parallel, flexible machine learning methods have emerged as particularly promising for enhancing predictive accuracy (Strydom et al., 2021). For example, Pichler et al. (2020) showed that machine learning models outperformed traditional regression approaches when predicting species interactions in plant-pollinator networks. Nevertheless, skepticism remains, and comparative approaches that evaluate predictive ability in different contexts are needed.

In this study, we aim to evaluate the relative importance of biotic and abiotic drivers of plant-pollinator interactions within seasonal dynamics using various analytical methods, and to assess the predictive power of species interactions under different analytical approaches and temporal scales (**Figure 1**). To achieve this, we collected data on plant-pollinator interactions, along with associated biotic (i.e., abundances, traits, and phenology) and abiotic variables (i.e., temperature), in three German botanical gardens. Our objective is to contribute to a mechanistic understanding of these drivers by using a semi-controlled environment with numerous co-flowering species that exhibit heterogeneous traits and diverse phylogenetic backgrounds. This setting allows us to test multiple potential combinations of pairwise interactions within the gardens and to explore in depth the role of traits and phylogenetic relatedness as surrogates for unmeasured traits.

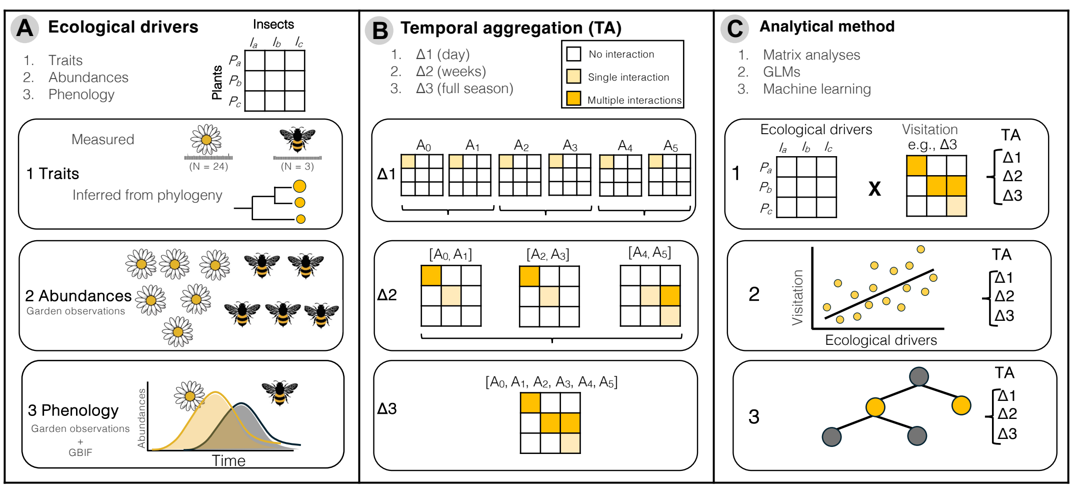

Figure 1. Workflow of the study, showing how different ecological drivers (A) are tested across various temporal scales (B) using different analytical methods (C). Panel A illustrates the main ecological drivers used in this study to evaluate plant-pollinator interaction patterns: (1) pollinator and floral traits, (2) pollinator and floral abundances, and (3) pollinator and floral phenology. Panel B presents the three levels of temporal aggregation considered in this study: (1) individual sampling days, (2) early, mid, and late season, and (3) the full flowering season. Panel C depicts the three main analytical approaches implemented in this study to

evaluate the effects of different ecological drivers across distinct levels of temporal aggregation: (1) matrix analyses, (2) generalized linear models (GLMs), and (3) machine learning

Methods

Study site and sampling method

Plant-pollinator interactions were monitored during the entire flowering season of 2023 in the botanical gardens of Halle, Jena, and Leipzig (Germany; **Figure S1**). These gardens belong to the PhenObs initiative (Nordt et al., 2021), a network of botanical gardens across the Northern Hemisphere that records phenological observations on a wide range of herbaceous species. The objective of PhenObs is to evaluate the impacts of climate change on species performance (e.g., Sporbert et al., 2022). In this study, plant-pollinator interactions were recorded for a subset of species from the PhenObs project within these three gardens.

Each garden was visited weekly, and plant-pollinator interactions were sampled for 15 minutes on each focal plant. On each sampling day, focal plants were surveyed in a random order, and only pollinator individuals contacting the reproductive organs of the plant were recorded as interacting species. In addition, to create a comprehensive inventory of pollinators and their interactions within the gardens, ten random sites were selected each sampling day, and plant-pollinator interactions were monitored for three minutes at each site, resulting in a total of 30 minutes of random observations per sampling day and garden.

Pollinator monitoring concluded once PhenObs species had largely ceased flowering in each garden, corresponding to the late phase of the flowering season. Random surveys nevertheless showed continued pollinator activity and flowering of non-PhenObs plants (**Figure S2**), indicating that sampling was constrained by focal plant phenology rather than pollinator availability. Consequently, two complementary sampling approaches were implemented: targeted observations on PhenObs plants, representing a subset of the available floral resources, and random stops within the gardens designed to characterise the broader pollinator assemblage that may not be captured through this subset.

Sampling effort and taxonomic coverage

This study monitored 78 focal plant species from the PhenObs initiative, encompassing 68 genera, 33 families, and 22 orders. Pollinator observations were conducted on 46 plant species in Halle (52.3 h), 43 in Jena (41.3 h), and 47 in Leipzig (49.8 h), with only 12 species shared among gardens. Complementary random census sampling expanded plant coverage, recording plant-pollinator interactions on 134, 124, and 125 plant species in Halle, Jena, and Leipzig, respectively (10.1, 11.3, and 10.4 h per garden), collectively capturing a broad taxonomic representation of flowering plants.

Focal observations captured a total of 175 pollinator species across the three gardens, with 123 species in Halle, 86 in Jena, and 72 in Leipzig. The random census sampling further enriched pollinator diversity, documenting an additional 68 species not observed on focal plants. Overall, the combined sampling approaches provided a comprehensive overview of the pollinator assemblages within these botanical gardens. The accumulation curves (**Figure S3**) indicate that sampling effort was sufficient to characterise pollinator assemblages associated with PhenObs plants across gardens. However, random observations proved more efficient for capturing overall pollinator diversity at the garden scale, providing a more comprehensive representation of the pollinator assemblage with lower sampling effort while covering a broader range of plant species.

Data description

To evaluate the different mechanisms that drive plant-pollinator interactions, detailed information on abundance, phenology and species traits for both plants and pollinators was recorded as follows:

(i) Floral abundance was annotated for each focal plant species, and given that botanical gardens are a semi-control setting with relatively low abundance of pollinators, we were also able to annotate the different number of pollinator individuals on each sampled plant when documenting interactions. Despite not being fully independent of interactions, this approach offers a more realistic estimate of pollinator abundance than the standard method of estimating abundances from interactions, which can bias our understanding of the mechanisms driving network structure (Blüthgen et al., 2006; Vizentin-Bugoni et al., 2014). Note that flowering species with tiny densely packed flowers were considered at inflorescence level (15.2% of the total flowering species considered in this study).

(ii) Given that plant phenology was already monitored on the PhenObs project, we have weekly phenological records with flowering percent for all focal plants. For 10 species with missing phenology data, we conducted generative additive models (GAM) with our records of flower number to build the phenology on the same sampling dates as the other species by using the *mgcv* package. Unlike plants, pollinator phenology relies on 'stochastic' interaction observations that may not represent their complete flying period. For this reason, we opted to complement our observational data from the Global Biodiversity Information Facility (GBIF). Germany is one of the countries with highest records of occurrence data (Rocha-Ortega et al., 2021), and in particular, their cities (e.g., botanical gardens) contain most of those records (Sweet et al., 2022). Thus, we opted to extract occurrence records around the three cities for the different pollinator species to complement the different pollinator phenologies. We first created a centroid within the 3 cities, and then, created a rectangular shape around it that expanded approximately 1000 kms in longitude and 500 kms in latitude (Figure S1). We used this shape because we expect environmental variables to be more consistent and phenology to be more similar across a longitudinal range than a latitudinal one. Thus, our download resulted in 2,933,772 occurrence records (https://doi.org/10.15468/dl.k5cnuf).

(iii) Plant reproductive traits were selected following the main traits driving the trait spectrum of flowering plants as shown in Lanuza et al. (2023) (i.e., flower number, plant height, style length, flower size, ovule number and selfing). In addition, we also included phenological traits (i.e., flower lifespan and flowering length), floral rewards (nectar volume and number of pollen grains per flower) and seed weight as a measurement of the reproductive investment (**Table S1a**). Pollinator traits included intertegular distance, body length and proboscis length (**Table S1b**). Both intertegular distance and body length were measured from images using ImageJ, and proboscis length was extrapolated for bee species by using the pollimetry package (Kendall et al., 2019). Non-bee species were complemented with proboscis values from the literature and images when possible.

*Consider phylogeny?

Analysess

To unravel the role of time and analytical methods in our understanding of the ecological drivers of plant-pollinator networks, this study analyses the role of abundance, phenology and traits at three different levels of temporal complexity (i.e., weekly, weekly aggregated and full season) using three different analytical approaches (see **Figure 1**). The three analytical approaches implemented are: (i) matrix analyses using Procrustes tests, (ii) generalized linear models (GLMs), and (iii) machine learning. Each of these methods provides complementary insights into the mechanisms driving plant-pollinator interactions across different temporal scales.

Matrix analyses

Visitation rate networks were built for each botanical garden for pollinators identified to species, and the main pollinator orders (Hymenoptera, Diptera, Coleoptera, Lepidoptera). Plant-pollinator interaction records were aggregated by garden and species pair to generate weighted bipartite interaction matrices. Interaction frequency (visitation rate) networks were calculated by dividing total interactions for each plant–pollinator pair by the total observation time for each plant species within a garden, thereby standardizing visitation rates. These standardized rates were summed across sampling occasions to produce garden-level visitation-rate networks representing cumulative seasonal interaction intensity while accounting for plant-specific sampling effort.

Abundance probability matrices were generated for each botanical garden using standardized relative abundances of pollinator species and floral resources. Pollinator and floral abundances were summed within each garden and converted to relative abundances by dividing species-level abundances by total community abundance. Finally, interaction probabilities were calculated by combining plant and pollinator relative abundances under an independence assumption, whereby the expected probability of interaction between a plant and a pollinator species was proportional to the product of their relative abundances.

Phenology probability matrices were calculated following a similar approach, but abundances were standardized within each species by dividing weekly observations by the total number of observations for that species. This procedure generates a temporal probability distribution describing the likelihood that a species is phenologically active in a given week. Weekly plant and pollinator probabilities were then combined under an independence assumption to estimate temporal interaction probabilities for each plant-pollinator pair.

Trait-based probability matrices were constructed using a size-matching approach between plant and pollinator traits. Pollinator intertegular distance and plant flower width were averaged at the species level to obtain a single representative trait value per species. Trait compatibility for each plant–pollinator pair was quantified using a Gaussian matching function, in which similarity declines smoothly with increasing trait mismatch and is controlled by a tolerance parameter. Pairwise matching scores were assembled into a global plant × pollinator matrix and subsequently subset for each botanical garden by aligning rows and columns to the species composition and ordering of each garden's interaction network and temporal scale.

First, matrix analyses were used to evaluate how similar interaction frequency matrix configurations were across temporal scales for each driver. This was assessed using Procrustes analysis implemented through the *protest* function, and the strength of configuration similarity was quantified using the Procrustes correlation coefficient (r). Analogous analyses were conducted using the Mantel test using the *mantel* function, and the strength of the association was quantified using the absolute values of the Mantel correlation coefficient. Because probability matrices for all drivers were scaled between 0 and 1, Euclidean distance was used as the distance metric in both analyses, as it preserves the geometric relationships among observations compared to alternative distance metrics.

Generalized linear models

Results

PROVIDE: Percentage of forbidden links due to phenological mistmatches by garden.

Figure 2

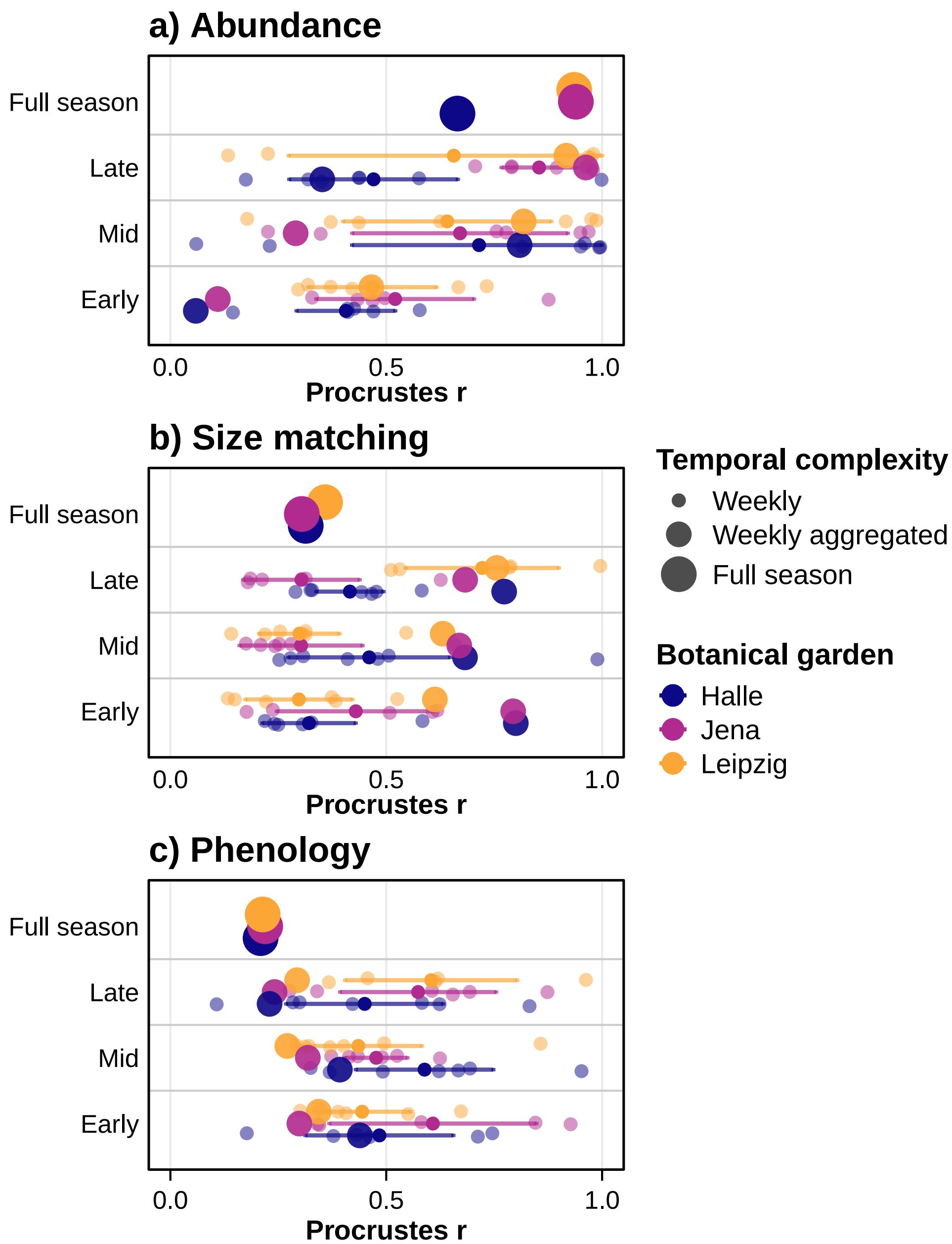

Figure 2. Procrustes association between visitation rate and abundance (a), phenology (b), and traits (size matching; c) across botanical gardens and three levels of temporal complexity (weekly, weekly aggregated and full season). Weekly aggregated values correspond to three seasonal categories (Early, Mid, and Late), obtained by pooling weekly networks into three equal temporal partitions of the full sampling period. Weekly values represent individual weekly networks and are displayed within the corresponding seasonal category. Because multiple weekly values are available per garden and seasonal category, weekly data are shown as individual points together with their mean and 95% confidence intervals (± 1.96 SE). Full-season values correspond to networks aggregated across the entire sampling period for each botanical garden. Temporal complexity is represented by point size, and botanical garden by colour.

Figure 3 GLm's

Figure 4 Machine learning

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Supporting information

Table S1 Summary of plant and pollinator traits

Traits	Mean	SD
a) Plant traits		
Autonomous selfing	6.45	21.67
Corolla diameter	26.22	22.09
Flower lifespan	8.08	8.38
Flowering length	50.48	26.62
Flowers per plant	550.97	1406.11
Nectar volume	1.33	3.80
Ovule number	66.68	92.00
Plant height	0.53	0.33
Pollen per flower	381413.84	1461347.93
Pollen size	42.10	28.81
Seed mass	0.01	0.04
Style length	9.48	13.25
b) Pollinator traits		
Body length	9.04	3.56
IT	3.05	1.37
Proboscis length	3.92	3.13

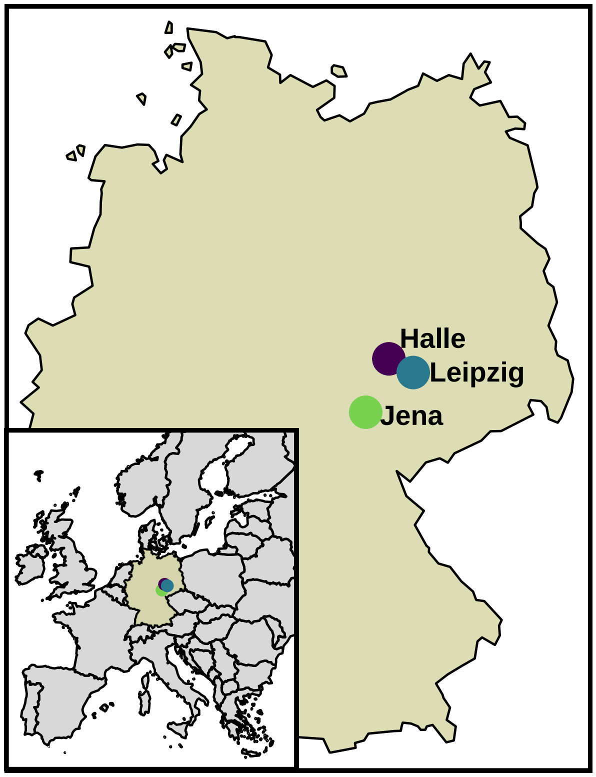

Figure S1.Map showing the locations of the three botanical gardens included in the study.

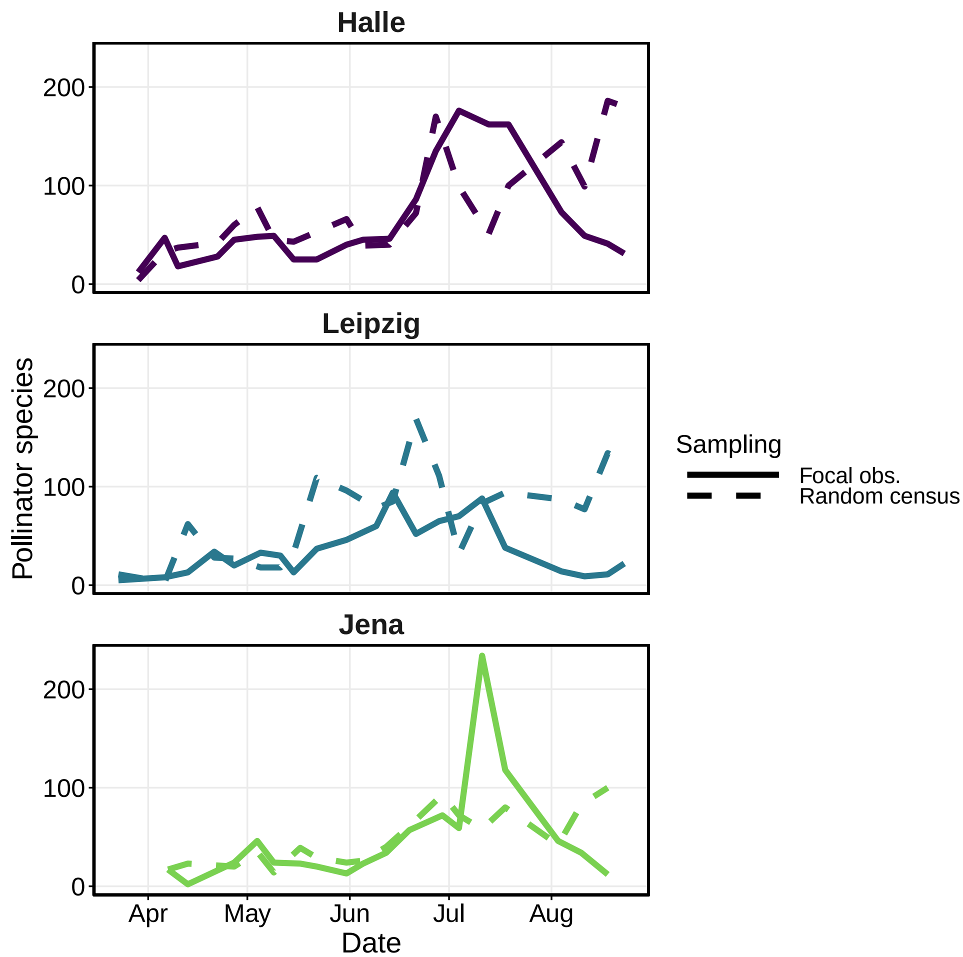

Figure S2. Temporal dynamics of plant-pollinator interactions across three botanical gardens. Solid lines represent focal observations on PhenObs plants, whereas dashed lines indicate random census observations across the gardens. Each panel corresponds to a different botanical garden: (a) Halle, (b) Leipzig, and (c) Jena.

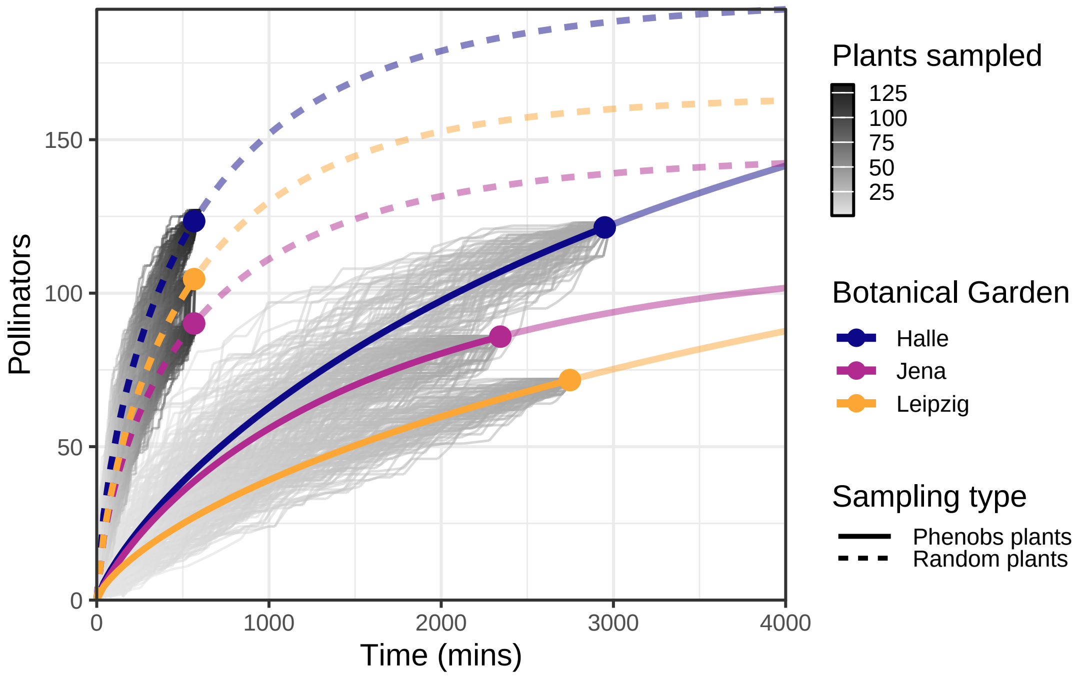

Figure S3. Accumulation curves of pollinator species across botanical gardens and sampling methods (dashed and solid lines). Each grey line represents one iteration of the accumulation curve, with a total of 100 iterations per garden and sampling method. The coloured lines represent the rarefied mean accumulation curves, with the dot indicating the total number of observed species. The segments of the lines extending beyond the dot represent the extrapolated number of expected species.

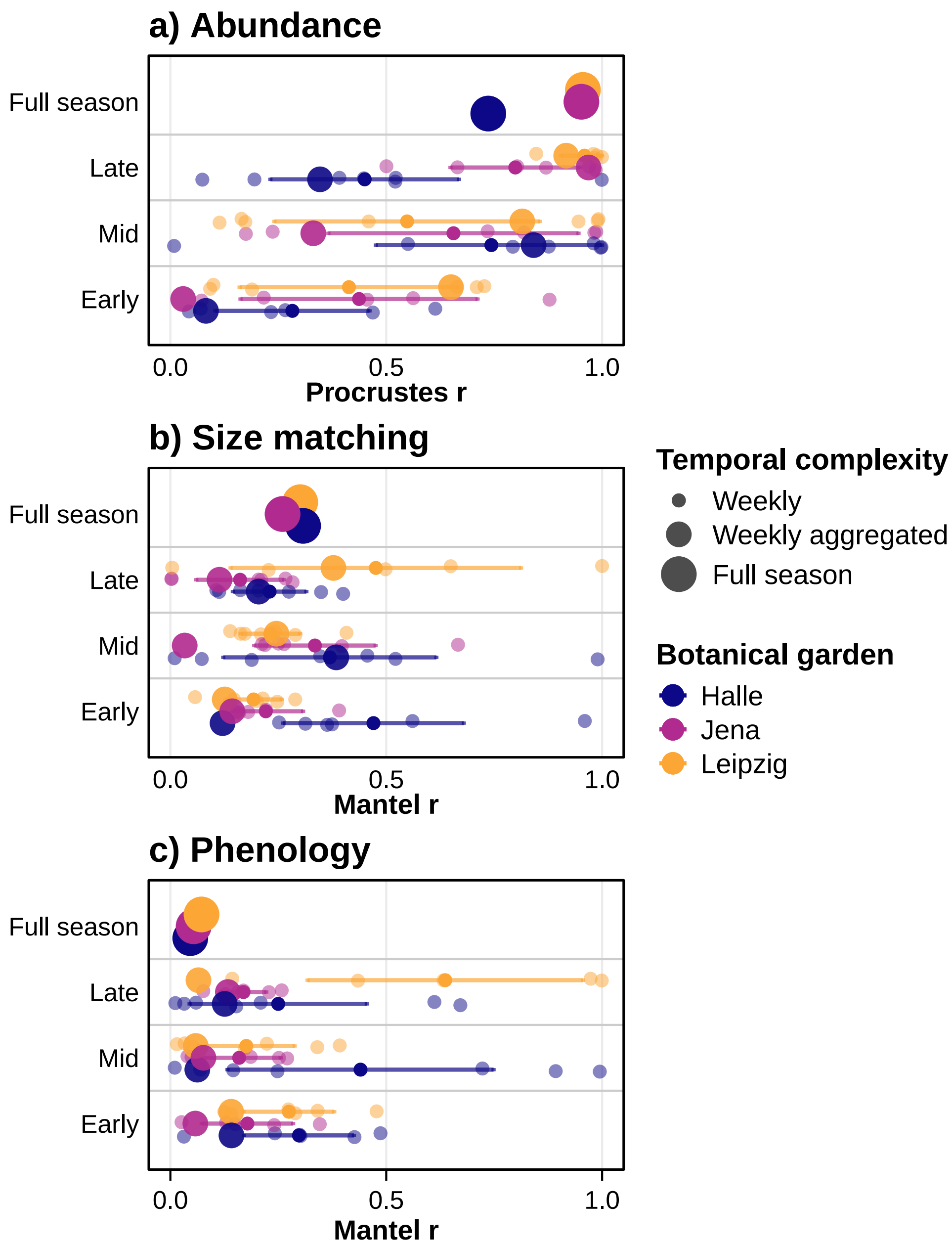

Figure S4. Mantel correlation between visitation rate and abundance (a), phenology (b), and traits (size matching; c) across botanical gardens and three levels of temporal complexity (weekly, weekly aggregated and full season). Weekly aggregated values correspond to three seasonal categories (Early, Mid, and Late), obtained by pooling weekly networks into three equal temporal partitions of the full sampling period. Weekly values represent individual weekly networks and are displayed within the corresponding seasonal category. Because multiple weekly values are available per garden and seasonal category, weekly data are shown as individual points together with their mean and 95% confidence intervals (± 1.96 SE). Full-season values correspond to networks aggregated across the entire sampling period for each botanical garden. Temporal complexity is represented by point size, and botanical garden by colour.